一，高维微分学

1，分片函数：$f(x,y)=\begin{cases} y-x+\dfrac{xy^2}{x^2+y^2} & (x,y)\neq(0,0) \\ 0 & (x,y)=(0,0) \end{cases}$

分析：(I)函数在原点的可微性，(II)函数在原点的方向导数。

解：(I)考察 $\lim\limits_{\Delta x,\Delta y\to 0}\dfrac{f(\Delta x,\Delta y)-f(0,0)-\Delta x f_x(0,0)-\Delta y f_y(0,0)}{\sqrt{(\Delta x)^2+(\Delta y)^2}}$

其中：$f_x(0,0)=\lim\limits_{\Delta y\to 0}\dfrac{f(\Delta x,0)-f(0,0)}{\Delta x}=\lim\limits_{\Delta x,\to 0}\dfrac{-\Delta x}{\Delta x}$=-1

$f_y(0,0)=\lim\limits_{\Delta y\to 0}\dfrac{f(0,\Delta y)-f(0,0)}{\Delta y}=\lim\limits_{\Delta x,\to 0}\dfrac{\Delta y}{\Delta y}$=1

$\lim\limits_{\Delta x,\Delta y\to 0}\dfrac{f(\Delta x,\Delta y)-f(0,0)-\Delta x f_x(0,0)-\Delta y f_y(0,0)}{\sqrt{(\Delta x)^2+(\Delta y)^2}}$

$=\lim\limits_{(x,y)\to(0,0)}\dfrac{xy^2}{(x^2+y^2)^{3/2}}$，设按照$x=ky$方向逼近，则容易知道

$\dfrac{xy^2}{(x^2+y^2)^{3/2}}=\dfrac{k}{(k^2+1)^{3/2}}$，与$k$相关，所以在原点不可微。

(II)，设$\vec{e}=\cos\theta\vec{i}+\sin\theta\vec{j}$，根据方向导数定义：

$\lim\limits_{r\to 0}\dfrac{f(r\cos\theta,r\sin\theta)}{r}=\dfrac{\partial f}{\partial e}(0,0)=\sin\theta-\cos\theta+\cos\theta\sin^2\theta$

2，考虑分片函数$f(x,y)=\begin{cases} \dfrac{1-e^{x(x^2+y^2)}}{x^2+y^2} & (x,y)\neq(0,0) \\ 0 & (x,y)=(0,0) \end{cases}$

(I)，获得$f(x,y)$在$(0,0)$点直到$o(r^4)$的展开式（设定符合展开的条件）

(II)，求$f(x,y)$的全部二阶偏导数和四阶偏导数

解:(I)直接对指数部分整体进行泰勒展开

$$f(x,y)=\frac{-xr^2-\frac{1}{2}x^2r^4+o(r^6)}{r^2}=-x-\frac{1}{2}x^2(x^2+y^2)+o(r^4)$$

$$(II)\ f(x,y)-f(0,0)=f_x x+f_y y+\frac{1}{2}(f_{xx}x^2+2f_{xy}xy+f_{yy}y^2)$$

$$+\cdots+\frac{1}{24}(f_{xxxx}x^4+4f_{xxxy}x^3y+6f_{xxyy}x^2y^2+4f_{xyyy}xy^3+f_{yyyy}y^4)+o(r^4)$$

对应展开式有:所有的二阶偏导数都是零

四阶偏导数: $f_{xxxx}(0,0)=-12,\ f_{xxyy}(0,0)=-2$

3，设四元数组$\{x,y,z,u\}$满足关系$u=f(x,y,z),g(e^y,z,x^2)=0,y=\cos x.$

式中的f,g都是$\mathscr{C}^1$函数

(I)设有因果分解：x为因/自变量,$\{y,z,u\}$为果/因变量，试讨论分解的充要条件

(II)，计算，y, z, u各自关于x的变化率/导数

(I)，若存在因果分解，则$D\begin{pmatrix}\begin{bmatrix}u-f(x,y,z)\\ g(e^y,z,x^2)\\ y-\cos x\end{bmatrix}\end{pmatrix}$非奇异

即:$\begin{bmatrix}-f_y & -f_z & 1\\ e^y g_z & g_z & 0\\ 1 & 0 & 0\end{bmatrix}=DF(y,z,u)=-g_z\neq 0$

$$(II)\ y_x=-\sin x, z_x=-\frac{\frac{\partial g}{\partial x}}{\frac{\partial g}{\partial z}}=-\frac{g_1e^y y_x+g_3 2x}{g_z},$$

$$u_x=f_x+f_y y_x+f_z z_x=f_x-f_y\sin x-f_z\frac{g_1e^y y_x+g_3 2x}{g_z}$$

二：高维积分学

$1,(I)$设E_{xy}是直线$y=0,x=1,y=x$围成的区域，计算:$\int_{E_{xy}}\sqrt{4x^2-y^2}dxdy$

(II)计算x=0, y=0,$\sqrt[4]{\frac{x}{a}}+\sqrt[4]{\frac{y}{b}}=1,a,b>0$所围面积

解（I)=$\int_0^1 dx \in [\ ] \int_0^x \sqrt{4x^2-y^2}dy$

设$y=2x\sin\theta$ $dy=2x\cos\theta d\theta, \theta \in \left[0,\frac{\pi}{6}\right]$

$I=\int_0^1 dx\int_0^{\frac{\pi}{6}} 4x^2\cos^2\theta d\theta = \int_0^1 4x^2 dx\int_0^{\frac{\pi}{6}}\cos^2\theta d\theta = \frac{\sqrt{3}}{6}+\frac{\pi}{9}$

(II)设$x=ar\cos^8\theta$ $y=br\sin^8\theta$ $\theta\in\left[0,\frac{\pi}{2}\right]$

则原区域为$r\in[0,1]$

$$\det J=\begin{vmatrix} a\cos^8\theta & -8ar\cos^7\theta\sin\theta \\ b\sin^8\theta & 8br\sin^7\theta\cos\theta \end{vmatrix}=8abr\cos^7\theta\sin^7\theta$$

故$S=\iint\limits_{E_{xy}} dxdy = 8ab\int_0^1 rdr\int_0^{\frac{\pi}{2}}\sin^7\theta\cos^7\theta d\theta$

$$=4ab\int_0^{\frac{\pi}{2}}\sin^7\theta\cos^6\theta d\sin\theta=\int_0^1 x^7\left(1-x^2\right)^3 dx=\frac{ab}{70}$$

2，令曲面Σ为:$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$,计算:

$\int_\Sigma \frac{dydz}{x}+\frac{dxdz}{y}+\frac{dxdy}{z}=\int_\Sigma\left(\frac{1}{x},\frac{1}{y},\frac{1}{z}\right)\bullet\vec{n}ds$,其中$\vec{n}$是外法向方向

解：设$\begin{cases} x=a\sin\theta\cos\varphi \\ y=b\sin\theta\sin\varphi \\ c\cos\theta \end{cases}$,则外法向为:$\begin{bmatrix} bc\sin^2\theta\cos\varphi \\ ac\sin^2\theta\sin\varphi \\ ab\sin\theta\cos\theta \end{bmatrix}$

$$I=\iint_{E_{\theta\varphi}}\frac{bc}{a}\sin\theta+\frac{ac}{b}\sin\theta+\frac{ab}{c}\sin\theta d\theta d\varphi=4\pi\left(\frac{bc}{a}+\frac{ac}{b}+\frac{ab}{c}\right)$$

3, 计算曲面积分: $\int_{\Sigma}\frac{xdydz+ydzdx+zdxdy}{\left(x^2+y^2+z^2\right)^{3/2}}=\int_{\Sigma}\frac{x\boldsymbol{i}+y\boldsymbol{j}+z\boldsymbol{k}}{\left(x^2+y^2+z^2\right)^{3/2}}\bullet\boldsymbol{n}dS,$

式中$\Sigma:\frac{x^2}{4}+\frac{y^2}{9}=1-z,\boldsymbol{n}$是外法向方向

解：注意到：$\vec{r}=\frac{x\boldsymbol{i}+y\boldsymbol{j}+z\boldsymbol{k}}{\left(x^2+y^2+z^2\right)^{3/2}},\nabla\bullet\vec{r}=\frac{3\left(x^2+y^2+z^2\right)^{3/2}-3\left(x^2+y^2+z^2\right)\left(x^2+y^2+z^2\right)^{1/2}}{\left(x^2+y^2+z^2\right)^3}=0$

考虑*Guass*公式:

$$\oint\limits_{\Sigma+\Sigma_1+\Sigma_\varepsilon}\vec{r}\bullet\vec{n}ds=\int\limits_V\nabla\bullet\vec{r}=0,\Sigma_1:\varepsilon<\frac{x^2}{4}+\frac{y^2}{9}\le 1,\Sigma_\varepsilon:x^2+y^2+z^2=\varepsilon^2,z>0,$$

首先计算:$\oint\limits_{\Sigma_1}\vec{r}\bullet\vec{n}ds\ \vec{n}=(0,0,1),\vec{r}=\frac{x\boldsymbol{i}+y\boldsymbol{j}}{\left(x^2+y^2+z^2\right)^{3/2}},\vec{r}\bullet\vec{n}=0$

$$\oint\limits_{\Sigma_\varepsilon}\vec{r}\bullet\vec{n}ds,\ \vec{n}=\frac{(x,y,z)}{\left(x^2+y^2+z^2\right)^{1/2}},\oint\limits_{\Sigma_\varepsilon}\vec{r}\bullet\vec{n}ds=\oint\limits_{\Sigma_\varepsilon}\frac{\left(x^2+y^2+z^2\right)}{\left(x^2+y^2+z^2\right)^2}ds$$

$$=\oint\limits_{\Sigma_\varepsilon}\frac{1}{\varepsilon^2}ds=2\pi\varepsilon^2\frac{1}{\varepsilon^2}=2\pi$$

4，计算曲线积分:$\int_C\left[(x+1)y^2+1\right]dx+2xydy+xy^2dz=\int_C\left\{\left[(x+1)y^2+1\right]\boldsymbol{i}+2xy\boldsymbol{j}+xy^2\boldsymbol{k}\right\}\bullet\boldsymbol{\tau}dl$

C是右半棱柱$|x|+|y|=a(y>0)$与平面$y=z$的交线从$(-a,0,0)$到$(a,0,0)$的部分，

$a>0$，走向为z轴正方向的右螺旋方向

解：考虑$\vec{F}(x,y,z)=\begin{bmatrix}(x+1)y^2+1\\2xy\\xy^2\end{bmatrix}\Rightarrow\nabla\times\vec{F}=\begin{bmatrix}2xy\\-y^2\\-2xy\end{bmatrix},$

又，所切得到平面外法向为$\vec{n}=\left(0,-\frac{\sqrt{2}}{2},\frac{\sqrt{2}}{2}\right)$

补上线段*NM*.运用*Stocks*公式

$$\int_{C\cup NM}\vec{F}(x,y,z)\bullet\boldsymbol{\tau}dl=\int_{\Delta MPN}\nabla\times\vec{F}\bullet\vec{n}dS$$

$$=\int_{\Delta MPN}\sqrt{2}xydS+\int_{\Delta MPN}\frac{\sqrt{2}}{2}y^2dS=0+\int_{\Delta MPN}\frac{\sqrt{2}}{2}y^2dS$$

$$\int_{\Delta MPN}\frac{\sqrt{2}}{2}y^2dS=2\int_0^a\left(ay^2-y^3\right)=\frac{a^4}{6}\Leftrightarrow\int_{C\cup NM}\vec{F}(x,y,z)\bullet\boldsymbol{\tau}dl=\frac{a^4}{6}$$

之后计算在*NM*上的线积分，容易得到$\int_{NM}\vec{F}(x,y,z)\bullet\boldsymbol{\tau}dl=\int_{NM}dl=2a$

故原式为$\frac{a^4}{6}$-2a

三，级数

1，分析级数:$\sum_{n=1}^{+\infty}\frac{(-1)^n}{\left[2n+(-1)^n\right]^{\alpha}}$的敛散性，需明确绝对收敛和条件收敛或发散的对应参数α范围

$$\frac{a_n}{a_{n+1}}=\begin{cases}-1,n\text{是偶数}\\ -\left(\frac{2n+3}{2n-1}\right)^{\alpha}=-\left(\frac{1+\frac{3}{2n}}{1-\frac{1}{2n}}\right)^{\alpha}=-\left(1+\frac{3\alpha}{2n}+o\left(\frac{1}{n^2}\right)\right)\left(1+\frac{\alpha}{2n}+o\left(\frac{1}{n^2}\right)\right)\\ =-\left(1+\frac{2\alpha}{n}+o\left(\frac{1}{n^2}\right)\right),n\text{是奇数}\end{cases}$$

$\alpha\le 0,$不绝对收敛，$\alpha>0,a_n\to 0$

$2\alpha>1\Rightarrow\alpha>\frac{1}{2},$绝对收敛，$\alpha\le\frac{1}{2},$不绝对收敛

$0<\alpha\le\frac{1}{2},$有$\frac{1}{\left(2n+(-1)^n\right)^{\alpha}}\downarrow$且趋于0,由$A-D$判别法知道条件收敛

$\alpha\le 0$，易知发散。

2，取得函数$f(x)=\ln\left(x+\sqrt{1+x^2}\right)$的幂级数表达，并且说明收敛域

解：$f'(x)=\sqrt{1+x^2}=\sum_{n=1}^{+\infty}\binom{-\frac{1}{2}}{n}x^{2n},$对其进行积分，则有：

$$f(x)=x+\sum_{n=1}^{+\infty}\binom{-\frac{1}{2}}{n}\frac{1}{2n+1}x^{2n+1},\text{系数}a_n=\binom{-\frac{1}{2}}{n}\frac{1}{2n+1}$$

$$\frac{a_n}{a_{n-1}}=-\frac{(n+1)(2n+3)}{\left(n-\frac{1}{2}\right)(2n-1)}\frac{1}{x^2}=-\frac{1}{x^2}\left(1+\frac{3}{2}n+o\left(\frac{1}{n^2}\right)\right)$$

$\frac{3}{2}>1,$故有收敛半径是1，考察原级数在1处和-1处均有收敛，故$[-1,1]$为收敛域

3，求幂级数$\sum_{n=1}^{+\infty}\frac{x^n}{n(n+1)}$和函数，并确定收敛域

收敛域：$\frac{a_n}{a_{n+1}}=\frac{1}{x}\frac{(n+1)(n+2)}{n(n+1)}=\frac{1}{x}\left(1+\frac{2}{n}\right)$，检验$x=1$知道绝对收敛

$S(x)=\sum_{n=1}^{+\infty}\frac{x^n}{n(n+1)}$，$f(x)=xS(x)$

$f'(x)=\sum_{n=1}^{+\infty}\frac{x^n}{n}$，$f''(x)=\sum_{n=1}^{+\infty}x^{n-1}=\frac{1}{1-x}$

积分即有：$S(x)=1+\left(\frac{1}{x}-1\right)\ln(1-x)$

注意$S(x)$的连续性，$S(x)=\begin{cases}1+\left(\frac{1}{x}-1\right)\ln(1-x) & x\in[-1,1),\text{且}x\neq 0\\ 1 & x=1\\ 0 & x=0\end{cases}$

4，判定函数项级数$\sum_{n=1}^{+\infty}(-1)^n\frac{x}{(1+x)^n}$在$x\geq 0$的一致收敛性

先考虑：$\varphi_n(x)=\frac{x}{(1+x)^n}$，　$n\to+\infty$时，$\varphi_n(x)\to 0$=：$\varphi(x)$

$\delta_n(x)=\varphi_n(x)-\varphi(x)=\frac{x}{(1+x)^n}$，$\delta_n(x)$在$x=\frac{1}{n-1}$取到最大值

故$\delta_n(x)<\frac{1}{n-1}\to 0$，$\varphi_n(x)\to 0$，且关于$n$单调，$\sum_{n=1}^{+\infty}(-1)^n$绝对值有界，

由$A-D$判别法知道原式在$x\geq 0$一致收敛